

Since

$$h(s) = \begin{cases} \frac{1}{\tau^2}s + \frac{2}{\tau}, & s \in [-2\tau, -\tau], \\ -\frac{1}{\tau^2}s, & s \in [-\tau, 0]. \end{cases} \quad (0.1)$$

Then,

$$\int_{-2\tau}^0 h(s)ds = 1$$

The model:

$$\begin{cases} \partial_t u = d_1 \Delta u + d_2 \nabla \cdot (u \nabla u_\tau) + f(u), & x \in \Omega, t > 0, \\ \frac{\partial u}{\partial n} = 0, & x \in \partial\Omega, t > 0, \end{cases} \quad (0.2)$$

where

$$u_\tau = \int_{-2\tau}^0 h(s)u(x, t+s)ds, \quad \tau \in \mathbb{R}_+ : (0, +\infty).$$

Throughout this paper, we assume that

$$(H_1) \text{ There exists } \bar{u} > 0 \text{ such that } f(\bar{u}) = 0, \text{ and } f_u = \partial_u f(u) < 0 \text{ at } \bar{u}.$$

which in order to make sure that $\bar{u}$ is locally asymptotically stable for the ODE associated with (1.7).

The linearization of (1.7) at $\bar{u}$ is

$$\partial_t u = d_1 \Delta u + d_2 \bar{u} \Delta u_\tau + f_u u. \quad (0.3)$$

The characteristic equation at $\bar{u}$ is

$$\lambda + d_1 \mu_n + d_2 \bar{u} \mu_n \int_{-2\tau}^0 h(s)e^{\lambda s} ds - f_u = 0, \quad (0.4)$$

or equivalently,

$$\lambda + d_1 \mu_n + \frac{1}{\lambda^2 \tau^2} (-2e^{-\lambda\tau} + e^{-2\lambda\tau} + 1) d_2 \bar{u} \mu_n - f_u = 0, \quad (0.5)$$

It can be seen that $\lambda = 0$ is not a root of (0.5). Thus, both sides of (0.5) can be multiplied by λ^2 to get

$$P_0(\lambda, \tau) + P_1(\lambda, \tau)e^{-\lambda\tau} + P_2(\lambda, \tau)e^{-2\lambda\tau} = 0, \quad (0.6)$$

where

$$\begin{aligned} P_0(\lambda, \tau) &= \lambda^3 + (d_1 \mu_n - f_u) \lambda^2 + \frac{1}{\tau^2} d_2 \bar{u} \mu_n, \\ P_1(\lambda, \tau) &= -\frac{2}{\tau^2} d_2 \bar{u} \mu_n, \\ P_2(\lambda, \tau) &= \frac{1}{\tau^2} d_2 \bar{u} \mu_n. \end{aligned}$$

1 Hopf bifurcation

Here we notice that the coefficients in $P_0(\lambda, \tau), P_1(\lambda, \tau), P_2(\lambda, \tau)$ are delay-dependent. However, we follow the method developed in [4] to determine critical parameter values of τ for Hopf bifurcation. The following items are true for $\tau \in \mathbb{R}_+, \omega \in \mathbb{R}_+$:

- (i) $\deg(P_0(\lambda, \tau)) \geq \max\{\deg(P_0(\lambda, \tau)), \deg(P_0(\lambda, \tau))\}$.
- (ii) $P_0(0, \tau_1) + P_1(0, \tau_1) + P_2(0, \tau_1) \neq 0$, that is, $\lambda = 0$ is not a characteristic root of (0.6);
- (iii) The polynomials $P_0(\lambda, \tau_1) + P_1(\lambda, \tau_1) + P_2(\lambda, \tau_1) \neq 0, \tau > 0$;
- (iv) $\lim_{Re\lambda \geq 0, |\lambda| \rightarrow \infty} \sup_{\tau \in I} \left\{ \left| \frac{P_1(\lambda, \tau_1)}{P_0(\lambda, \tau_1)} \right| + \left| \frac{P_2(\lambda, \tau_1)}{P_0(\lambda, \tau_1)} \right| \right\} < 1$;
- (v) For each τ , $F(\omega, \tau) = 0$ has finite real roots;
- (vi) Each positive root $\omega(\tau)$ of $F(\omega, \tau) = 0$ is continuous and differentiable at τ whenever it exists.

For each τ , (1.3) has at most a finite number of real roots, which ensures that there might have only a finite number of gates for the roots to cross the imaginary axis.

To obtain the stability criterion of $\bar{u}$. Let $\lambda = i\omega (\omega > 0)$, we have

$$\begin{aligned} \omega^3 \sin \omega \tau + \left[\frac{2}{\tau^2} d_2 \bar{u} \mu_n - (d_1 \mu_n - f_u) \omega^2 \right] \cos \omega \tau &= \frac{2}{\tau^2} d_2 \bar{u} \mu_n, \\ -\omega^3 \cos \omega \tau - (d_1 \mu_n - f_u) \omega^2 \sin \omega \tau &= 0. \end{aligned} \quad (1.1)$$

Then, we have

$$\begin{aligned} \cos \omega \tau &= \frac{\frac{2}{\tau^2} d_2 \bar{u} \mu_n (d_1 \mu_n - f_u)}{\left[\frac{2}{\tau^2} d_2 \bar{u} \mu_n - (d_1 \mu_n - f_u) \omega^2 \right] (d_1 \mu_n - f_u) - \omega^4}, \\ \sin \omega \tau &= - \frac{\frac{2}{\tau^2} d_2 \bar{u} \mu_n \omega}{\left[\frac{2}{\tau^2} d_2 \bar{u} \mu_n - (d_1 \mu_n - f_u) \omega^2 \right] (d_1 \mu_n - f_u) - \omega^4}. \end{aligned} \quad (1.2)$$

According to $\sin^2 \omega \tau + \cos^2 \omega \tau = 1$, we have

$$F(\omega, \tau) = \omega^6 + A_2 \omega^4 + A_1 \omega^2 + A_0 = 0. \quad (1.3)$$

where

$$\begin{aligned} A_2 &= 2(d_1 \mu_n - f_u)^2, \\ A_1 &= [(d_1 \mu_n - f_u)^3 - \frac{4}{\tau^2} d_2 \bar{u} \mu_n] (d_1 \mu_n - f_u), \\ A_0 &= -\frac{4}{\tau^2} d_2 \bar{u} \mu_n [(d_1 \mu_n - f_u)^3 + \frac{1}{\tau^2} d_2 \bar{u} \mu_n]. \end{aligned} \quad (1.4)$$

Now, $F(\omega, \tau)$ determines ω in terms of τ . Finding solutions of (1.3) is equivalent to looking for the positive solutions of

$$G(\omega, \tau) = z^3 + A_2 z^2 + A_1 z + A_0 = 0 = 0. \quad (1.5)$$

For (1.5), we have $A_2 = 2(d_1 \mu_n - f_u)^2 > 0$. Then, by Descartes' rule of signs, we can obtain that the polynomial equation (1.3) has at most one positive real root.

Let $I = \{\tau : \tau > 0 \text{ and } \omega(\tau) \text{ is a positive root of (1.3)}\}$. Then, if $\tau \notin I$, there are no positive solution of (1.3). Now, for any $\tau \in I$ (where $\omega(\tau)$ is a positive root of (1.3)), we can define $\theta(\tau) \in [0, 2\pi]$ as the solution of (1.3):

$$\begin{aligned} \cos \theta(\tau) &= \frac{\frac{2}{\tau^2} d_2 \bar{u} \mu_n (d_1 \mu_n - f_u)}{[\frac{2}{\tau^2} d_2 \bar{u} \mu_n - (d_1 \mu_n - f_u) \omega^2] (d_1 \mu_n - f_u) - \omega^4} = \frac{T(\omega(\tau), \tau)}{R(\omega(\tau), \tau)}, \\ \sin \theta(\tau) &= -\frac{\frac{2}{\tau^2} d_2 \bar{u} \mu_n \omega}{[\frac{2}{\tau^2} d_2 \bar{u} \mu_n - (d_1 \mu_n - f_u) \omega^2] (d_1 \mu_n - f_u) - \omega^4} = \frac{S(\omega(\tau), \tau)}{R(\omega(\tau), \tau)}. \end{aligned} \quad (1.6)$$

Substituting $\omega = \omega(\tau)$ in (1.6), $\theta(\tau) \in [0, 2\pi]$ can be determined as follows:

$$\theta(\tau) = \begin{cases} \arctan \frac{T(\omega(\tau), \tau)}{S(\omega(\tau), \tau)}, & \text{if } \sin \theta > 0, \cos \theta > 0, \\ \frac{\pi}{2}, & \text{if } \sin \theta = 1, \cos \theta = 0, \\ \pi + \arctan \frac{-T(\omega(\tau), \tau)}{S(\omega(\tau), \tau)}, & \text{if } \cos \theta < 0, \\ \frac{3\pi}{2}, & \text{if } \sin \theta = -1, \cos \theta = 0, \\ 2\pi + \arctan \frac{T(\omega(\tau), \tau)}{S(\omega(\tau), \tau)}, & \text{if } \sin \theta < 0, \cos \theta > 0, \end{cases} \quad (1.7)$$

Here we notice that for $\tau \in I$, $\theta(\tau)$ defined above is continuous at τ . Furthermore if $\theta(\tau) \in (0, 2\pi)$, $\tau \in I$, then $\theta(\tau)$ is also differentiable at τ . Now, the relation between the arguments " $\theta(\tau)$ " in (1.6) and " $\omega(\tau)\tau$ " in (1.3) for $\tau \in I$ must be

$$\omega(\tau)\tau = \theta(\tau) + 2k\pi, n \in \mathbb{N}_0.$$

Hence, we can define the maps $\tau_k : I \rightarrow \mathbb{R}_{+0}$ given by

$$\tau_k(\tau) := \frac{\theta(\tau) + 2k\pi}{\omega(\tau)}, n \in \mathbb{N}_0, \tau \in I.$$

where $\omega(\tau)$ is a positive simple root of $F(\omega, \tau) = 0$. Let us introduce the functions $I \rightarrow \mathbb{R}$

$$S_k(\tau) := \tau - \tau_k(\tau), \tau \in I, n \in \mathbb{N}_0, \quad (1.8)$$

that are continuous and differentiable at τ . We notice that the values of $\tau(\in I)$ at which stability switches may occur, are the solutions of $S_k(\tau) = 0$ for some $n \in \mathbb{N}_0$ provided the transversality condition is satisfied. To find out the transversality condition, we differentiate the characteristic equation (0.6). Then, we obtain

$$\frac{d\lambda}{d\tau} = -\frac{d_2 \bar{u} \mu_n [-\frac{2}{\tau^3} + (\frac{4}{\tau^3} + \frac{\lambda}{\tau^2}) e^{-\lambda\tau} - (\frac{2}{\tau^3} + \frac{2\lambda}{\tau^2}) e^{-2\lambda\tau}]}{3\lambda^2 + 2(d_1 \mu_n - f_u)\lambda + \frac{2}{\tau} d_2 \bar{u} \mu_n (e^{-\lambda\tau} - e^{-2\lambda\tau})} \quad (1.9)$$

Let $\lambda = i\omega$, then

$$\begin{aligned} \frac{d\lambda}{d\tau}|_{\lambda=i\omega} &= -\frac{d_2\bar{u}\mu_n[-\frac{2}{\tau^3}e^{i\omega\tau} + (\frac{4}{\tau^3} + \frac{i\omega}{\tau^2}) - (\frac{2}{\tau^3} + \frac{2i\omega}{\tau^2})e^{-i\omega\tau}]}{[-3\omega^2 + 2(d_1\mu_n - f_u)i\omega]e^{i\omega\tau} + \frac{2}{\tau}d_2\bar{u}\mu_n(1 - e^{-i\omega\tau})} \\ &= -\frac{C + Di}{A + Bi}. \end{aligned} \quad (1.10)$$

where

$$\begin{aligned} C &= \frac{d_2\bar{u}\mu_n}{\tau^2}(-\frac{4}{\tau}T + \frac{4}{\tau}R - 2\omega S), \\ D &= \frac{d_2\bar{u}\mu_n}{\tau^2}(\omega R - 2\omega T), \\ A &= (-3\omega^2 - \frac{2}{\tau}d_2\bar{u}\mu_n)T - 2(d_1\mu_n - f_u)\omega S + \frac{2}{\tau}d_2\bar{u}\mu_n R, \\ B &= 2(d_1\mu_n - f_u)\omega T + (-3\omega^2 - \frac{2}{\tau}d_2\bar{u}\mu_n)S. \end{aligned}$$

Since

$$\text{sign}\{\frac{d\text{Re}\lambda}{d\tau}|_{\lambda=i\omega}\} = \text{sign}\{\text{Re}(\frac{d\lambda}{d\tau}|_{\lambda=i\omega})\}$$

we compute

$$\frac{d\text{Re}\lambda}{d\tau}|_{\lambda=i\omega} = -\frac{AC + BD}{A^2 + B^2}.$$

from which

$$\text{sign}\{\text{Re}(\frac{d\lambda}{d\tau}|_{\lambda=i\omega})\} = \text{sign}\{-AC - BD\}.$$

Then we have the following theorem.

Theorem 1.1. *Let $\omega(\tau)$ be a positive root of (1.3) defined for $\tau \in I$, and at some $\tau^* \in I$,*

$$S_k(\tau^*) = 0$$

for some $n \in \mathbb{N}_0$. Then a pair of simple conjugate pure imaginary roots $\lambda_+(\tau^) = i\omega(\tau^*)$ and $\lambda_-(\tau^*) = -i\omega(\tau^*)$ of (0.6) exists at $\tau = \tau^*$ which crosses the imaginary axis from left to right if $\delta(\tau)^* > 0$ and crosses the imaginary axis from right to left if $\delta(\tau^*) < 0$, where*

$$\delta(\tau^*) = \text{sign}\{\frac{d\text{Re}\lambda}{d\tau}|_{\lambda=i\omega(\tau^*)}\} \quad (1.11)$$

2 Normal Form of Hopf Bifurcation

In the remaining of this section, we further compute the Hopf bifurcation properties where $\Omega = (0, l\pi)$. Notice that $\mu_n = (\frac{n}{l})^2$, each being simple, and a corresponding eigenfunction is $\varphi_n(x) = \cos \nu_n x$ with $\nu_n = \sqrt{\mu_n}$. Assume that $\lambda = \pm i\omega_n$ is a pair of purely imaginary roots of characteristic equation (0.4) when $\tau = \tau_{n,j}$, $j \in \mathbb{N}_0$ for some fixed $n \geq 1$.